Noah Sun

(647) 700-5886 | nn2sun@uwaterloo.ca | sunnoah.ca | github.com/LegendaryArk | /in/sunnoah

Education

University of Waterloo – BASc. in Mechatronics Engineering (Candidate) — September 2025 – April 2030
Relevant Coursework: Algorithms and Data Structures, Circuits, Digital Logic — GPA: 3.92 / 4.00

Experience

Embedded Flight Software Developer – Waterloo Aerial Robotics Group — September 2025 – Current

- Led 4 engineers to develop a 32 MB flash driver using C++ on the STM32 Nucleo to store config details and crash logs
- Built RTOS-integrated Flash Translation Layer with garbage collection and wear leveling, enhancing flash longevity

V5RC Team Programming Lead – Checkmate Robotics | Demo Video | GitHub — June 2019 – August 2025

- Led team of 6 in robot design and library architecture, winning the Create (innovation) Award at VEX Worlds 2024
- Engineered ArkLib, a modular C++ motion control library focused on scalability and flexibility using OOP, qualifying the team to VEX Worlds in 5 consecutive years and winning the Think (programming) Award at international events
- Designed and tuned PID and control algorithms, achieving 1”/1° precision movements across 300+ matches

Vex AI Competition Team Embedded Systems Lead – Mi3L Schools — May 2024 – June 2024

- Developed an API for the VEX Brain to process 3D spatial data from the Intel RealSense depth camera via the Jetson Nano for adaptive field navigation, winning the Innovate Award and Skills World Champion at VEX AI Worlds 2024

Forward Deployed Engineer – Ascendance Foundry — January 2026 – April 2026

- Built a GTM automation that uses Python Playwright to find leads and generate outreach emails using Claude API
- Developed 5 AI workflow automations, introducing input validation and reducing human error by 80%
- Implemented 2 ETL pipelines using Azure Data Factory, delivering per-project statistic dashboards in Power BI

Co-founder and Lead Developer – Elapse | Google Play | App Store | GitHub — June 2024 – August 2025

- Led 4 developers to build an iOS/Android Flutter app serving the 100k+ member VEX Robotics Competition community with real-time schedules, rankings, and match data, receiving 3500+ downloads and a 5.0-star rating
- Designed adaptive match times and scouting forms, increasing team productivity at tournaments by 35%
- Integrated Firebase and Cloud Functions to sync data between team members, reducing scouting times by 30%

Projects

AutoFleet | GitHub

- Built a multi-robot central arbiter with space-time A* path planning to avoid collisions during coordinated traversal
- Programmed ESP32 WiFi-UART bridge with a TCP server to relay JSON commands, enabling wireless robot control
- Integrated AprilTag localization with EMA-smoothed pose fused with wheel odometry, boosting precision by 30%

Decks - Card Deck Shuffler | GitHub

- Architected a full embedded system for an automated playing card shuffler built on a custom 3D-printed chassis
- Developed embedded firmware in MicroPython on Raspberry Pi Pico for motor control, sensor polling, and status LED

Autonomous Robot Navigation System | Demo Video | GitHub

- Implemented A* path planning on a LiDAR-generated occupancy grid with cost-weighted traversal, dynamic replanning, and distance-weighted obstacle inflation at 0.1 m resolution
- Built Pure Pursuit controller with odometry-fused map memory for persistent global mapping across robot motion

Technical Skills

Languages: C/C++, Python, Java, JavaScript, TypeScript, Dart, MicroPython
Embedded/Hardware: Raspberry Pi, STM32, NVIDIA Jetson, Arduino, UART, SPI, Soldering, SolidWorks, 3D Printing
Robotics/Controls: ROS2, RTOS, PID Control, Motor Control, Sensor Fusion, Kinematics, Foxglove
AI/Software: OpenCV, TensorFlow, FastAPI, Express.js, Node.js, React Native, Flutter, Matplotlib
Data/DevOps: PostgreSQL, MySQL, Microsoft Azure, Firebase, Pandas, Git, Docker, Linux, GitHub Actions, REST APIs